

Multivariable Calculus (MT1008)

Sessional-1

Date: 25th February, 2026

Total Time: 1 Hours

Course Instructor(s)

Total Marks: 50

Dr. Akhlaq Ahmad Bhatti (Moderator)

Total Questions: 5

Dr. Mazhar Hussain

Dr. Nasir Ali

Dr. M. Aziz

Dr. Imran Nadeem

Ms. Laila Yawar

Ms. Eesha Mir

Dr. Hina Dilawar

Important Instruction: 1. Only simple calculator is allowed. Use of programmable calculator or any helpful devices will result F grade in the course.

2. Question # 1(e) must be returned to the invigilator along with answer sheet.

CLO #1: Defining Functions of Several Variables, Computing Partial Derivatives, Directional derivative and gradient vectors, Tangent planes and Extreme value(s).

Question # 1(a)[10]: Show that

$$f(x,y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

is continuous at every point except the origin.

Question # 1(b)[10]: If $w = \ln(x^2 + y^2 + z^2)$, $x = u e^v \sin u$, $y = u e^v \cos u$, $z = u e^v$.

Evaluate the partial derivative $\partial w / \partial u$ and $\partial w / \partial v$ at point $(u,v) = (-2,0)$. Also draw the dependency tree diagram.

An interior point of the domain of a function $f(x,y)$ where both f_x & f_y are zero or where one or both of f_x and f_y do not exist is a critical point of f .

A differentiable function $f(x,y)$ has a saddle point at a critical point (a,b) if in every open disk centered at (a,b) there are domain points (x,y) where $f(x,y) > f(a,b)$ and domain points (x,y) where $f(x,y) < f(a,b)$. The corresponding point $(a,b,f(a,b))$ on the surface $z = f(x,y)$ is called a saddle point of the surface.

Question # 1(c)[10]: Find the linearization $L(x, y, z)$ of $f(x, y) = xy^2 + y \cos(x - 1)$ at the point $(x_0, y_0) = (1, 2)$. Find an upper bound for the error incurred in replacing f by L on the rectangular region

$$R: |x - 1| \leq 0.1, |y - 2| \leq 0.1.$$

Question # 1(d)[10]: Define critical and saddle points. Find the absolute maximum and minimum values of $f(x, y) = 2 + 2x + 4y - x^2 - y^2$ on the triangular region in the first quadrant bounded by the lines $x = 0$, $y = 0$, and $y = 9 - x$.

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$f(x, y) \approx L(x, y)$$

is the standard linear approximation of f at (x_0, y_0)



$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$, satisfies the inequality

$$|E(x, y)| \leq \frac{1}{2} M (|x - x_0| + |y - y_0|)^2$$

Total differential of f :

$$df = f_x(x_0, y_0) dx + f_y(x_0, y_0) dy$$

$$|E| \leq \frac{1}{2} M (|x - x_0| + |y - y_0| + |z - z_0|)^2$$